FINM3013 Introduction to Financial Derivatives

Chapter 03

Solution to Class exercises

1. B. As a general rule the futures maturity month should be as close as possible to but after the month when the asset will be purchased. In this case the asset will be purchased in June and so the best contract is the July contract.
2. A. The user of the commodity takes a long futures position. The gain on the futures is 63.50−59 or $4.50. The effective paid realized is therefore 64−4.50 or $59.50. This can also be calculated as the March 1 futures price (=59) plus the basis on July 1 (=0.50).
3. D. To reduce the beta by 0.3 we need to short 0.3×36,000,000/(900×250) or 48 contracts.
4. A. Index futures can be used to remove the impact of the performance of the overall market on the portfolio. If the market is expected to do well hedging against the performance of the market is not appropriate. Hedging cannot correct for a poorly diversified portfolio.
5. A. The price received by the trader is the futures price plus the basis. It follows that the trader's position improves when the basis increases.